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Overview

Plantir is an Internet of Things (IoT) plant care solution. It collects environmental data via a handful of sensors and sends it to an app, where a user can monitor and water their plant. Plantir ensures a plant remains healthy when a user is not home or simply forgets to water their plant. The design uses an application that allows users to view environmental metrics in real time, water their plants, and view and import predefined care parameters for a wide variety of plant species.

Goals

- Enable users to better care for their plants by providing:
 - live plant metrics
 - automatic and manual watering control
 - ideal conditions for each plant species
- Create an enclosure to house relevant hardware
- Build a real-world product and show technical depth by having:
 - robust security
 - an intuitive user interface
 - a reliable system
- Allow for autonomous maintenance of plants
- Develop a performant, native Android application
- Design and allow for modular prints
- Keep the total cost below \$100

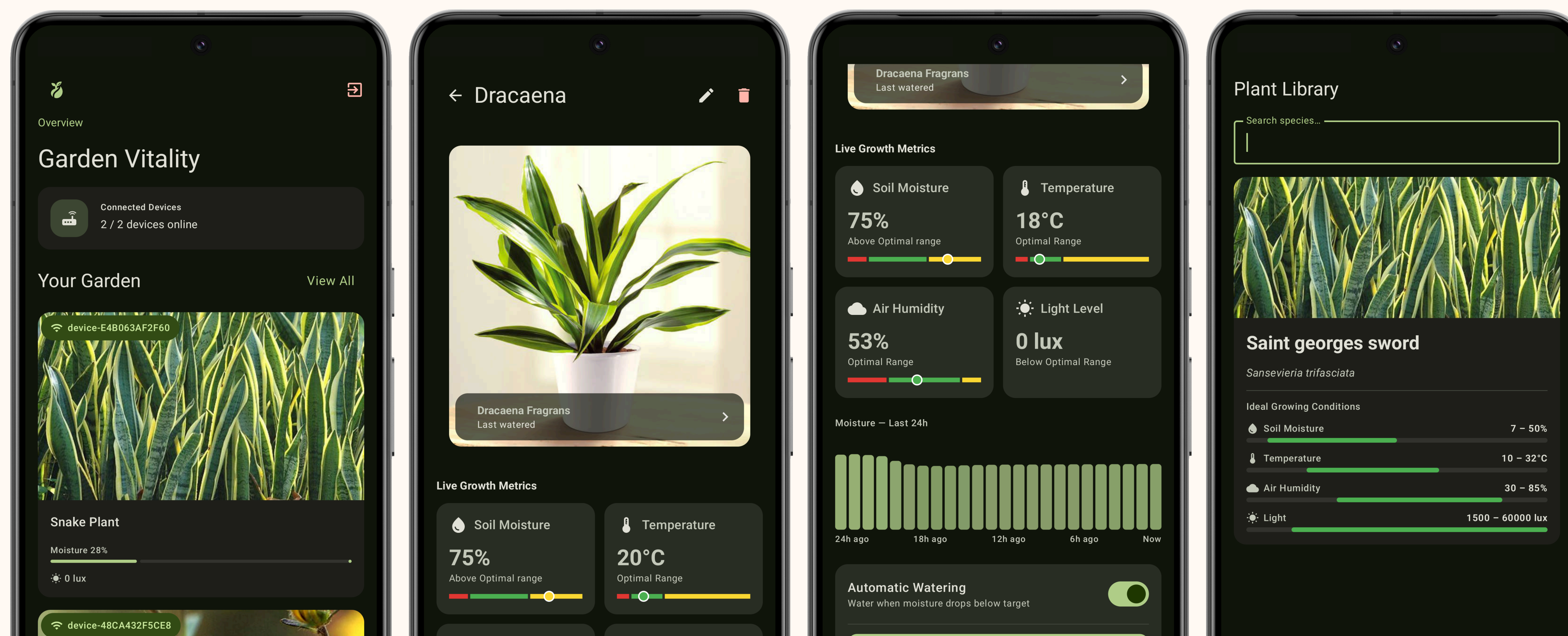
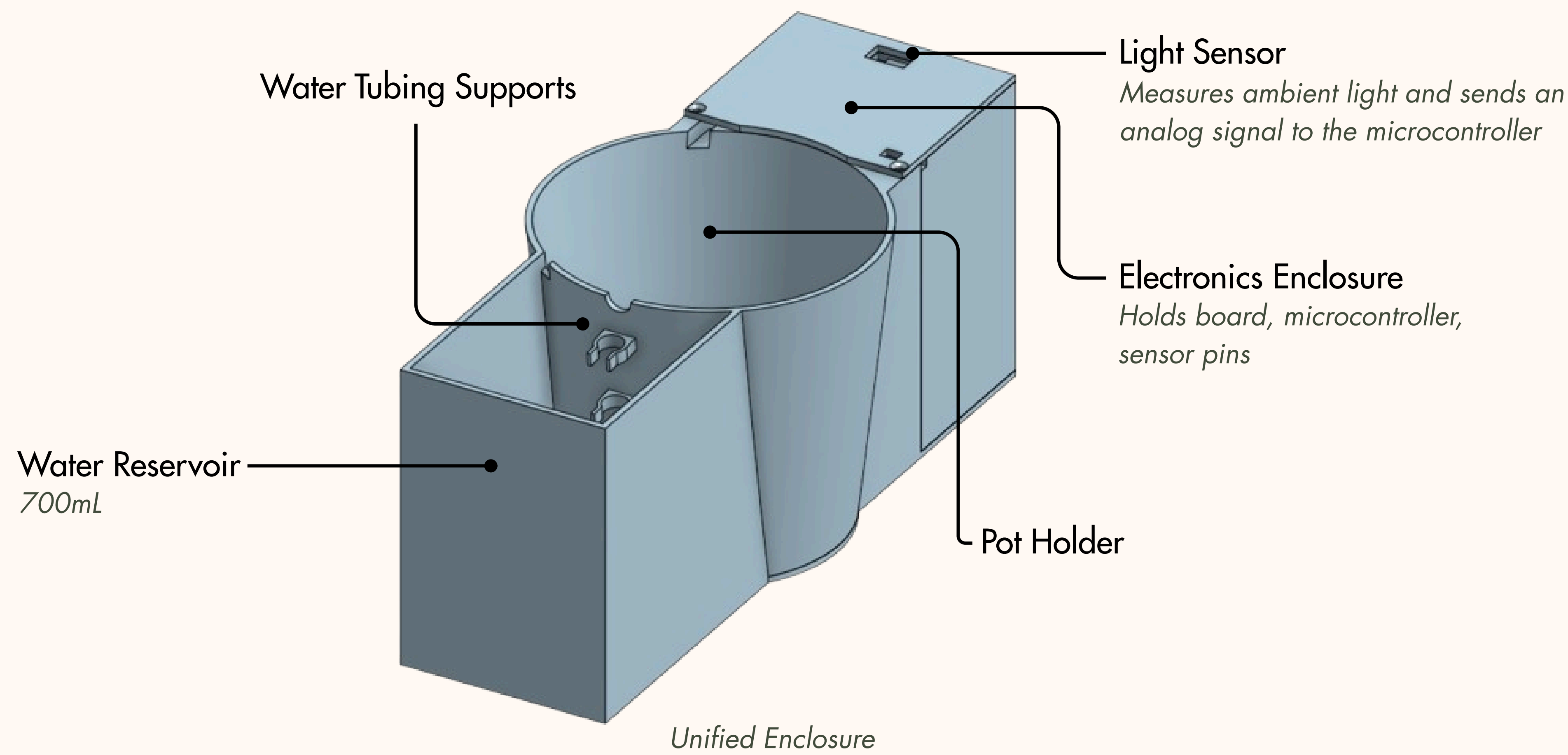
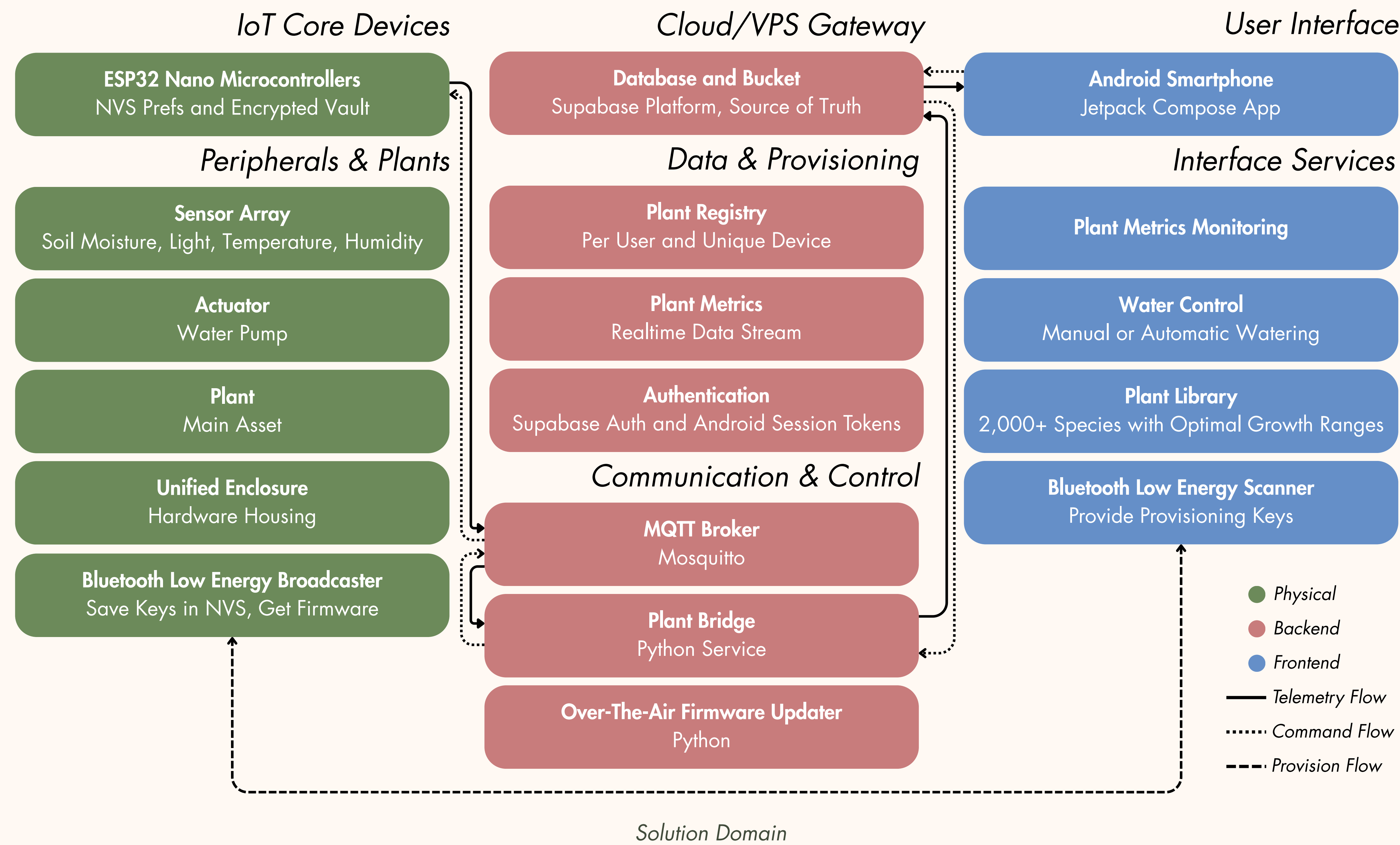
Challenges

- Creating a design where the end-user can assemble and set up their device with minimal technical knowledge
- Connecting all aspects of our hardware/software configuration to send/receive data and commands
- Keeping data secure through communication channels and servers
- Making a robust and reliable workflow that handles orphaned data and missing hardware
- Learning many new technologies and protocols

PLANTIR



IoT Plant Care



Technologies



Constraints

- Cost – \$50 in sensors and hardware, optionally \$30/mo for backend services
- Security – communication of keys and user data
- Partially network reliant – loss of internet is loss of metrics
- User resources – access to a 3D printer is ideal
- Time – Limited project hours
- Shared hardware authentication – controllers authenticate to MQTT with shared node credentials
- Initial firmware – the ESP32 needs initial firmware for BLE provisioning

Achievements

- Manual and automatic plant watering
- Live plant metrics via an Android application
- Zero-touch provisioning flow that allows non-technical users to easily set up plant monitoring using Bluetooth Low Energy (BLE)
- Updates are sent Over-The-Air (OTA) to devices with fleet management
- Credentials are secured by vaulting into Non-Volatile Storage (NVS) on-device
- Production-grade security with Access Control Lists (ACL) and Row-Level Security (RLS)
- Clean, simple, and unified enclosure that holds a plant, water reservoir, actuator, and sensors

Future Work

- Create both a DIY and a shelf-ready hardware kit
- Create printable models for use with different-sized or different-shaped pots
- Add the ability to control lighting fixtures
- Design an application for Apple devices
- Add network migration for devices
- Plant care plans tailored to each plant species
- Integrate more types of sensors (pH, nutrient-density)
- Dynamic authentication for devices